

Hacettepe University | Computer Engineering CMP 720 EMBEDDED SYSTEMS DESIGN

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Midterm Project Peer Evaluation Form

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Instructions

For each project, briefly summarize the work and provide your evaluation **individually (not as a group)**. Your response should reflect your understanding of the problem, methodology, and evaluation plan, and include critical insights, strengths, limitations, and possible improvements. Skip your own group's evaluation and simply state that it is your group.

Students are required to evaluate only the projects presented within their own class group (Hacettepe or ASELSAN).

Guiding points (address in one coherent paragraph):

- What problem is being solved, why is it important, and what are the most important embedded system design considerations for this problem (e.g., timing, reliability, energy, safety, security, scalability, etc.)?
- What is the main idea of the proposed methodology?
- What do you think are the strengths and limitations of the approach?
- Is there anything you would improve or do differently?
- What is the most critical or challenging component?
- Is the evaluation plan sufficient? What, if anything, is missing?

Group 1

Umut Kalman, Eren Karadağ

Project Title: *Context-Aware Deadline Scheduling Under Memory Contention for Embedded Real-Time Systems*

This project tackles memory contention in mixed-criticality systems, which causes dangerous jitter. The solution is a lightweight FreeRTOS wrapper using a slack-aware heuristic to defer memory-heavy tasks. Its primary strength is minimizing memory thrashing without kernel modifications, guaranteeing safety via EDF fallback. However, it statically profiles memory intensity, ignoring dynamic intra-task variations. Improving the $O(n)$ queue scanning to $O(1)$ would enhance scalability. The most challenging aspect is bounding the scheduler's overhead so it doesn't negate the time saved. The evaluation plan is highly sufficient, validating STM32L452 hardware metrics (jitter, overhead) against a theoretical offline ACO baseline.

Group 2

Yusuf Çağrı Ögüt

Project Title: *Hybrid Cellular-LoRa Network Based Agriculture Monitoring and Control Application*

This project enables off-grid hobby garden irrigation using a hybrid Cellular SMS and local LoRa network. Its core strength is extreme energy efficiency (1 mAh/day), using latched valves and 99 percent deep-sleep cycles for 100+ days of autonomy. A limitation is the total reliance on manual SMS commands. Implementing autonomous local irrigation based on sensor thresholds would improve this. The main challenge is balancing ultra-low power constraints with reliable underground radio transmission. The evaluation plan is thorough, using STM32/SIM800L hardware and oscilloscopes to empirically validate sub-60s latency, packet delivery ratios, and actual power draw.

Group 3

Yusuf Tahir Orhan

Project Title: *Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm*

This is my group.

Group 4

Mehmet Yusuf Sezgi, Rana Elif Albayrak

Project Title: *Real-Time CAN Decoder*

This project tackles real-time CAN bus decoding, where dense, high-frequency vehicle traffic (250 μ s per frame) causes reception latency and data loss. The methodology employs an STM32 MCU with Zephyr RTOS, utilizing a two-stage RAM buffering strategy to absorb traffic bursts. Its primary strength is decoupling rapid CAN reception from slow SD card writes, preventing frame drops while acting strictly as a secure, passive monitor. A limitation is the reliance on experimentally inferred signal meanings due to lacking an official DBC file. Improving this could involve integrating lightweight offline machine learning to auto-classify unidentified signals. The most critical challenge is resolving shared SPI bus contention between the CAN receiver and SD card. The evaluation plan is thorough, using DWT cycle counters to measure pipeline timing and Renode simulations to stress-test buffer capacity under heavy loads.

Group 5

Metin Irkıcıatal, Anıl Arda

Project Title: *Edge-AI Based Smart Access Control System*

ASELSAN group - Not evaluated.

Group 6

Oğulcan Uğuroğlu, Barış Büyükyılmaz

Project Title: *Contention-Aware Duplication Scheduling for NoC Systems*

This project tackles classical task duplication in DAG scheduling, which often ignores Network-on-Chip (NoC) contention and inadvertently exacerbates congestion. The methodology introduces a contention-aware heuristic within a list-based scheduler, tentatively evaluating duplication candidates based on Earliest Finish Time (EFT) improvements. Its main strength is selectively avoiding unnecessary task replication, thereby preserving shared network bandwidth. A limitation is its strict reliance on static, offline scheduling with deterministic routing. Upgrading the system to adapt dynamically to runtime contention would significantly improve robustness. The primary challenge is accurately estimating complex NoC contention during tentative scheduling without introducing massive computational overhead. The evaluation plan is highly sufficient, simulating synthetic DAGs on 2D mesh NoCs under varying communication-to-computation ratios (CCR) to rigorously measure makespan and network utilization.

Group 7

Ahmet Yiğitalp Demirci

Project Title: *Real-Time Energy-Constrained UAV Optimization*

ASELSAN group - Not evaluated.

Group 8

Sami Enes Yılmaz, Kadir Emre Avcı

Project Title: *SmartAquaria System*

ASELSAN group - Not evaluated.

Group 9

Mehmethan Ayrım

Project Title: *Energy-Aware Mixed-Criticality Scheduling*

ASELSAN group - Not evaluated.

Group 10

İrem Gül Subaşı

Project Title: *Fault Injection and Resilience Profiling of Lightweight Transformers*

ASELSAN group - Not evaluated.

Group 11

Süleyman Ege Karakaya

Project Title: *Adaptive Data-Movement Strategy for Crypto Accelerators*

ASELSAN group - Not evaluated.

Group 12

Gökçer Erol

Project Title: *Acoustic Localization System for Search and Rescue*

This project tackles survivor localization in disaster environments where traditional sensing fails due to rubble. The methodology deploys ESP32-S3 nodes, synchronized via GNSS PPS, to detect acoustics using Edge AI and transmit timestamp metadata via LoRa to a Jetson Nano gateway for TDoA multilateration. Its main strength is minimizing bandwidth by transmitting only timestamps instead of raw audio, coupled with robust noise filtering. A limitation is the single-gateway bottleneck. Implementing a LoRa mesh-routing protocol would improve resilience. The most critical challenge is maintaining strict microsecond-level ($\pm 10\mu\text{s}$) clock synchronization across distributed nodes. The evaluation plan is highly comprehensive, utilizing a 50x50m physical field with both Line-of-Sight and Non-Line-of-Sight rubble simulations to empirically measure RMSE localization accuracy, latency, and clock drift.

Group 13

Emre Doğan, Yavuz Selim Kızıltaş

Project Title: *Optimization of Edge AI Pipeline on TDA4VM*

ASELSAN group - Not evaluated.

Group 14

Mehmet Yusuf Bilge, Sezil Ağırbaş

Project Title: *Closed-Loop Vestibular Stimulation System*

This project tackles fatal falls caused by vestibular dysfunction by developing a wearable, closed-loop Galvanic Vestibular Stimulation (GVS) system. The methodology uses an STM32F411 and FreeRTOS, combining Complementary and Kalman filters for tilt estimation to drive a PID-controlled current pump. Its main strength is a defense-in-depth safety architecture, including a hardware-level 2.1mA clamp and watchdog timers, ensuring absolute electrical safety. A limitation is the reliance on single-axis tilt estimation; integrating multi-axis data or gait detection would improve real-world efficacy. The most critical challenge is balancing a strict 15ms end-to-end latency with multi-layered safety overhead. The evaluation plan is highly sufficient, utilizing motorized tilt simulations and physical fault injections to empirically validate sway reduction, control jitter, and sub-1ms safety shutdowns.

Group 15

Ayberk Aygün, Berkay Taşkın

Project Title: *Quadrotor Simulation on NVIDIA Jetson*

This project addresses the computational bottleneck of evaluating quadrotor controller robustness, where sequential CPU-based Monte Carlo simulations are too slow for real-time self-tuning. The methodology partitions workloads on an NVIDIA Jetson: the CPU executes a strict 5ms cascaded PID control loop, while the GPU concurrently runs parallel Monte Carlo trajectory rollouts to evaluate and update optimal gains. Its primary strength is exploiting the Jetson's unified memory to eliminate PCIe latency, enabling rapid mid-flight tuning. However, a limitation is its purely simulation-based nature; physical deployment would introduce unmodeled aerodynamic complexities. The most critical challenge is preventing shared memory contention between the CPU and thousands of GPU threads without violating the 5ms control deadline. The evaluation plan is comprehensive, benchmarking CPU vs. GPU speedups and assessing control metrics under mass, inertia, and noise perturbations.

Group 16

Yasin Arslan, Süleyman Sertaç Üst

Project Title: *Real-Time Audio Pipeline Analysis on Heterogeneous SoCs*

ASELSAN group - Not evaluated.

Group 17

Oğuz Kağan Yağlıoğlu, Emre Eşef Kılıçdoğan

Project Title: *Predictive Maintenance of BLDC Motors via Multi-Sensor Fusion*

This project tackles predictive motor maintenance to prevent costly industrial downtime. The methodology proposes an ultra-low-power ($\leq 10\mu A$) STM32WLE5 edge node that extracts load-invariant vibration and acoustic features for a condition-aware Random Forest classifier. Its core strength is pragmatic engineering: explicitly feeding load and speed data into the model to boost cross-load accuracy to 92 percent, and smartly merging ambiguous faults into an uncertain fault class to prioritize high recall over precision. A limitation is the current reliance on a pre-recorded public dataset. Implementing an unsupervised autoencoder could further help detect entirely novel anomalies. The main challenge is closing the domain gap between the pristine lab dataset and their upcoming custom PCB deployment. The evaluation plan is highly thorough, benchmarking multiple ML models using F1-scores and outlining a Phase 2 physical validation on their own BLDC motor using LoRaWAN.

Group 18

Hüseyin Eren Doğan

Project Title: *EdgeRAG: Offline RAG on Embedded Systems*

ASELSAN group - Not evaluated.

Group 19

Ali Bölücü

Project Title: *Spectral-Seeded NSGA-II for Many-Core Systems*

This project tackles the NP-hard problem of multi-application mapping on Many-Core NoCs, where standard NSGA-II algorithms suffer from slow convergence due to random initialization. The methodology proposes a Spectral-Seeded NSGA-II, utilizing Fiedler vector analysis to cluster highly communicative tasks and injecting them as "elite seeds" into the initial population. Its primary strength is drastically accelerating convergence while simultaneously optimizing communication costs and thermal distribution. A limitation is its strict reliance on static, design-time mapping, completely ignoring dynamic runtime workload fluctuations. The most critical challenge is carefully balancing the elite-to-random seed ratio to prevent premature convergence to local optima. The evaluation plan is highly sufficient, using MATLAB's PlatEMO framework to rigorously benchmark a 73-task merged workload against random NSGA-II and pure static spectral baselines.

Group 20

İlter Doğan Dönmez

Project Title: *Autonomous Obstacle Avoidance Vehicle using LiDAR*

This project addresses the need for cost-effective autonomous vehicle testing by proposing a low-power, miniature obstacle avoidance system. The methodology involves emulating a LiDAR-based navigation pipeline in Renode, utilizing an RTOS to filter 360-degree point clouds into simplified directional weights for a 5-stage motor control loop. Its primary strength is achieving reliable indoor navigation in low-light environments without computationally expensive computer vision. A key limitation is a physical blind spot for obstacles shorter than the LiDAR's mounting height. Integrating a downward-facing ultrasonic sensor could easily resolve this. The most critical challenge is processing 8,000 LiDAR data points per second under strict real-time deadlines without overflowing the MCU's limited memory. The evaluation plan is sufficient, using Renode to inject artificial sensor data via GPIO manipulation to stress-test the control logic before physical deployment.